INCH-POUND
A-A-59576A
10 July 2017
SUPERSEDING
A-A-59576
20 July 2004

COMMERCIAL ITEM DESCRIPTION

TOW BAR, AIRCRAFT TOWING, KC-10 AIRCRAFT

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers the KC-10 tow bar, which is used for towing and steering the aircraft by attaching it to the aircraft's nose landing gear (NLG) and being pulled by a tow tractor. It will be used to move aircraft to various locations around the flightline without powering up the aircraft, and to egress and regress aircraft from hanger facilities. The tow bar, once attached to the aircraft NLG strut, will then be attached to powered tugs to facilitate movement on the flightline.

2. SALIENT CHARACTERISTICS.

2.1 Functional requirements. The tow bar shall be suitable for its intended use. Structural elements shall be rigid and possess the requisite strength. Individual components shall be durable, reliable, and suited for their intended purpose. Service and parts shall be readily available for all components. The tow bar must be able to pull an aircraft weighing 593,000 pounds (lbs) (268,900 kilograms) without any deformation.

2.2 Interface. The interface requirements shall be as specified in table I.

TABLE I. Interface requirements.

Overall Length	129.10 Inches Nominal
Special Features	32.00 Inches in Height; 34.00 Inches in Width; 480 Lbs
Towed-end Connection Type	Lunette Eye
Towing-end Connection Type	Hook With Locking Pin
Wheel Quantity	Two (2)
Length Adjustability	Non-adjustable

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AMSC N/A FSC 1730

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2.3 Standard commercial unit. The tow bar unit shall, as a minimum, be in accordance with the requirements of this CID and shall be the contractor's standard commercial product. Additional or better features, which are not specifically prohibited by this CID, but which are a part of the contractor's standard commercial product, shall be included in the tow bar being furnished. The use of standard parts shall be maximized where they suit the purpose.

2.4 Materials. Selection of materials, processes, and parts utilized in the design and construction of the tow bar shall meet the best commercial practices and quality, unless otherwise noted, and shall be commensurate with the overall operational and performance requirements specified herein. The materials shall be of uniform quality and condition, free from any defects that may adversely affect appearance, strength, endurance, wear resistance, and/or serviceability. Parts that are subject to corrosion in a marine environment shall be aluminum flame sprayed, zinc galvanized, or constructed of stainless steel, brass, bronze, or other corrosion-resistant material. Unless protected against electrolytic corrosion, dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other.

2.5 Design. The tow bar body consists of a cylindrical body. At one end of the tow bar is a lunette eye to which the tow tractor connects. At the opposite end of the tow bar is a mechanical head assembly to which the aircraft's front landing gear connects. The tow bar shall be equipped with a wheeled undercarriage to serve as the means to make vertical height adjustments while attaching the end connection. The undercarriage shall be extendable by the use of either a mechanical device or a hydraulic pump and cylinder arrangement. If a hydraulic arrangement is selected, the undercarriage shall retract automatically by spring or other mechanism upon release of hydraulic pressure. The wheels, as well as any protruding stands, shall clear level ground by a minimum of 4 inches (in) during towing operations. The unit shall permit easy accessibility for maintenance and service. The construction shall be such as to prevent conditions hazardous to personnel or detrimental to equipment.

2.6 Safety. The tow bar shall have an axial-load limiter that fails at 83,700 lbs drawbar pull. The load limiter shall be a shear-pin type device equipped with a shear pin designed to break under excessive load. The shear pin will break before damage to the aircraft can occur. Shear pin failure is indicated by a sudden jerk of the tow vehicle and by an audible snap. The tow bar shall also have a load-limiter device to limit torsional loads on the NLG strut to 350,000 inch-pounds. The load limiter shall be a shear-pin type device. Upon overload of the axial or torsional shear pin, the load shall be transferred to retaining bolts that prevent separation of the aircraft from the towing vehicle. Following a shear pin failure, the tow bar must be taken out of service for shear pin replacement. As a safety feature, the tow bar shall have a minimum factor of safety of 3.0 against permanent deformation at the loads applied above.

2.7 Storage. On the tow bar, storage shall be provided for a spare set of shear pins for each shear pin device. The tow bar shall be delivered with a full set of spares in place.

2.8 Corrosion resistance. Materials used in the construction/design must be corrosion resistant to support worldwide deployment scenarios. Corrosion/deterioration prevention and control for the tow bar shall be in accordance with best commercial practices.

2.9 Finish. Unless specified in the contract, all painting and finishing shall be in accordance with the best commercial practice in the industry. All painted surfaces shall be color number 26173, gray, per SAE-AMS-STD-595 and top coat per MIL-PRF-85285, type I. All finishing processes

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utilized in the design and construction of the unit shall result in a durable, non-fading, long-lasting finish. Finish warranty shall be equivalent to best commercial warranty offered or a minimum of 3 years, whichever is greater. Warranty is not to include damage caused by common maintenance activities.

2.10 Maintenance. Use of standard hand tools enhances maintainability and decreases maintainer's cost. The tow bar shall be field maintainable using a minimum number of common, standard hand tools or other equipment. No more than 24 common hand tools and no more than 10 special tools shall be required to maintain the tow bar. Special tools require approval of procuring activity and shall be funded in conjunction with the basic end-item contract. One set of special tools will be required per location. Metric tools are not considered special tools. All major components shall be replaceable by field-level maintenance technicians.

2.11 Instructions and nameplate. Complete instructions on the use and maintenance of the tow bar shall be included. A corrosion-resistant nameplate shall be securely attached to the unit. The nameplate shall include the following:

Nomenclature
Manufacturer's name/CAGE code
Manufacturer's serial number
Manufacturer's part number
Date of manufacture
Contract number
National Stock Number (NSN)
Warranty information

2.12 Construction. The tow bar shall be constructed of parts that are free of harmful or extraneous materials. All edges shall be either rounded or beveled, unless sharpness is required to perform a necessary function. All materials used in the construction of the unit shall be compatible in a manner that will in no way prevent compliance with the performance requirements set forth in this CID.

2.13 Workmanship. Workmanship of the tow bar equipment shall be of a high quality and shall be commensurate with the requirements specified herein. The manufacturer shall provide any or all accessories that are necessary to make the unit complete and workable as standard.

2.14 Operation. The tow bar shall be completely operable by one operator. The tow bar shall easily operate between the full ranges of heights.

2.15 Transportability. The unit shall be as small as practical and consistent with meeting the specified performance characteristics as stated herein. The unit shall be transportable by airlift, spring-ride truck or trailer, rail car, boat, or ship. The unit shall meet all transport aircraft ramp-negotiation requirements specified in MIL-HDBK-1784.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, or biobased materials. The contractor is encouraged to use recycled, recovered materials, or biobased products to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), part 23.403.

4. PRODUCT CONFORMANCE PROVISIONS.

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4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance. Minor changes to the design will be considered by the procuring activity provided that significant performance, reliability, durability, and safety characteristics of the design are retained.

4.2 Bid sample. When specified in the contract or purchase order, a bid sample shall be provided for government verification of compliance with the following inspection(s). The design strength calculations and/or tests of the load-limiter devices, retaining bolts, and body assembly shall be provided by the tow bar manufacturer.

4.3 Examination. The tow bar shall be visually examined to assure that it meets the requirements of this CID.

5. PERFORMANCE REQUIREMENTS.

5.1 Performance. The tow bar shall demonstrate that it meets the performance requirements of this CID.

6. PACKAGING REQUIREMENTS.

6.1 Packaging. Preservation, packaging, and marking shall be as specified in the contract or purchase order.

7. NOTES.

7.1 Addresses for obtaining copies of referenced documents.

7.1.1 Federal Acquisition Regulations (FAR). FARs may be obtained from the Superintendent of Documents, Government Printing Office, Washington, DC 20402, or online at https://www.acquisition.gov/?q=browsefar.

7.1.2 Government documents. Copies of specifications, standards, and handbooks are available from DLA Document Services, 700 Robbins Avenue, Building 4/D, Philadelphia, PA 19111-5094, or online at https://assist.dla.mil.

7.2 Known acceptable products. Known acceptable products are as identified in table II.

TABLE II. Known acceptable products.

NSN	Part Number	CAGE (Manufacturer)
1730-01-465-0729	215030-1	08118 (Stanley Aviation Corporation)

7.3 Ordering data. The contract or order should specify the following:

CID document number and revision
Packaging requirements

7.4 Commercial products. As part of the market analysis and research effort, information was gathered from the following vendors of commercial/industrial products. At the time of CID preparation and coordination, these vendors were known to have commercial/industrial

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products that meet the requirements of this CID. NOTE: This information should not be considered as a list of approved vendors or be used to restrict procurement to only the vendors shown here.

Stanley Aviation Corporation
2501 Dallas St.
Aurora, CO 80010-1009
Cage: 08118

7.5 Subject term (key word) listing.

Aircraft ground movement
Aircraft ground parking
Aircraft ground handling

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MILITARY INTEREST

Custodian:
Air Force – 184

Preparing Activity:
Air Force – 184

Agent:
Air Force – 99

(Project 1730-2017-006)

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